

Sonal Panda

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SUMMARY

Final-year B.Tech student in Computer Science (AIML specialization) with strong programming and AI/ML skills. Experienced in deep learning, reinforcement learning and ML development. Proven problem-solver with hands-on projects, internships, and research experience, passionate about leveraging technology for impactful solutions.

EDUCATION

SRM Institute of Science and Technology
B.Tech in Computer Science (AIML Specialization)
GPA: 9.19 / 10

Sep 2022 – Present
Chennai, Tamil Nadu, India

Mothers Public School
Class 12 (CBSE), PCM
Percentage: 82%

2020 – 2022
Bhubaneswar, Odisha

Loyola School
Class 10 (ICSE)
Percentage: 91.4%

2020
Bhubaneswar, Odisha

WORK EXPERIENCE

Research Intern, DRDO CABS LAB, Bangalore

Dec 2023 – Jan 2024

- Applied satellite image analysis for precise object detection in defense projects.
- Implemented REDET algorithm and PCA with deep learning for enhanced reconnaissance.
- Collaborated within a multidisciplinary research team on defense-grade data processing.

Research Intern, IIT Bhubaneswar

May 2024 – July 2024

- Developed decentralized applications using blockchain (MetaMask, Sepolia test network).
- Implemented smart contracts ensuring secure and transparent transactions.

Intern, OpenRAG.in

May 2025 – July 2025

- Contributed to RAG-based LLM development for context-aware AI responses.
- Applied agentic AI to enable autonomous, goal-driven reasoning.
- Beta-tested the Android mobile application, providing performance feedback.

PROJECTS

Episodes-Based Traffic Signal Control: A Deep Reinforcement Learning Approach

- Developed AI-driven traffic optimization using **Deep Q-Networks** and fluid-dynamic simulation (**Taichi**).
- Modeled real-time traffic as fluid density; trained agents to minimize congestion adaptively.
- Visualized performance improvement via heatmaps and animations.
- *Technologies: Python, TensorFlow, Taichi, Reinforcement Learning*

Comparative Study of Adam vs PSO Optimizers for DRL-Based Traffic Control

- Conducted analysis comparing **Adam** and **Particle Swarm Optimization** in training DQN agents.
- Simulated intersection traffic balancing queue minimization and throughput maximization.
- **Adam outperformed PSO** in efficiency and reward optimization.
- *Technologies: Python, TensorFlow, DRL, PSO, Adam*

Content-Adaptive Watermarking for Generative Video with Adversarially Trained Detector

- Built a robust **DCT-based watermarking** and **CNN detector** pipeline resilient to attacks.
- Integrated adversarial attacks, temporal GRU detection, and **ECC (Hamming(7,4))** for robustness.
- Improved metrics from TPR=0.80/FPR=0.80 to **TPR=0.90/FPR=0.00**.
- *Technologies: PyTorch, OpenCV, GRU, ECC, DCT, Adversarial Training*

PUBLICATIONS

- “Episodes-Based Traffic Signal Control: A Deep Reinforcement Learning Approach with Fluid-Dynamic Simulation,” *ICITSM 2025*, 28–29 April 2025, Tamil Nadu, India. (Volume I, 8 pages)
- “A Comprehensive Comparative Study of Adam and Particle Swarm Optimization for Deep Reinforcement Learning-Based Traffic Signal Control,” *ICMEET 2025 (Springer Conference Proceedings)*, London, UK. (21 pages) [Publication in Progress]

AWARDS & ACKNOWLEDGEMENTS

Best Paper Award for ICMEET–UK’25

Dean of SRM’s Appreciation and acknowledgement for supporting role in Enrollment Process of 2023-2024; 2024-2025

Certificate of Acknowledgment for serving as Committee Head in Aaruush (SRM’s National Level Techno-management Fest)

SKILLS

Technical Skills

Programming: C, C++, Java, Python, MATLAB, SQL, PL/SQL, HTML/CSS

Tools Technologies: Git, GitHub, Blockchain Basics, Data Structures and Algorithms, Object-Oriented Programming (OOP)

Operating Systems: Windows, Linux

Soft Skills

Public Speaking, Leadership, Communication, Management, Agile Methodology

CO-CURRICULAR ACTIVITIES

Presented papers at international conferences:

- ICITSM 2025, Tamil Nadu, India
- ICMEET 2025 (Springer Conference Proceedings), London, UK

Participated in multiple hackathons and technical workshops

EXTRA-CURRICULAR ACTIVITIES

Member of Computational Intelligence Club, contributing to projects and events

Committee Head at national-level techno-management fest **AARUUSH**, SRMIST

Volunteered at **Chetana Blind School**, mentoring and assisting children